Report 2: Genomic characterization of the 2026 Bundibugyo virus disease outbreak in the Democratic Republic of the Congo

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This analysis includes 989 Bundibugyo virus (BDBV) genomes sampled through October 5, 2026 and uploaded through October 19; report prepared October 28, 2026.

Main findings

- The genomic epidemic size appears to have broadly stabilized since early June 2026; there is no longer an apparent decline in late August 2026. This is likely due to the additional genomic diversity provided by genomes from outside of Mongbwalu, Bunia, and Rwampara.
- The stabilization of the effective population size therefore likely reflects the averaging of a declining outbreak in these health zones with growing outbreaks in other health zones. This would broadly corroborate the epidemiology, with cases in the rest of Ituri and Nord Kivu proportionally increasing through August and September 2026, whereas the daily number of cases in Mongbwalu, Bunia, and Rwampara appear to be decreasing in the same time period. We note that this is a preliminary finding and multiple explanations need to be investigated.
- There is an increase in health zones with confirmed cases but no genomes, including Biena, Mutwanga, and Manguredjipa.
- Although sublineages are broadly not structured within Ituri, the Mongbwalu BDBV genomes appear to have some structure in the phylogeny, suggesting there could be multiple patterns of movement in the outbreak.
- The intensifying outbreak in Nord Kivu is focused in Katwa, Beni, Butembo, Mabalako, Oicha, and Musienene health zone but limited genomic sampling means understanding how these are linked remains a challenge. Mandima in Ituri is currently undersampled considering its recent surge in cases in September and genomes from there only available through mid-August.

The latest genomic epidemiology situation and analysis is available on the INRB dashboard: https://inrb-umie.github.io/BDBV2026-Epidemic_Dashboard/genomic-epidemiology.html

Table 1: Key quantitative parameters.

Parameter	Value
Genomes generated/analysed	989 generated; 982 after QC (767 Ituri, 10 Nord Kivu, 1 Sud Kivu)
Median genome coverage	97.1% (IQR 92.0–97.2)
Evolutionary rate	1.1e-03 subs/site/year (95% HPD 1.0e-03–1.2e-03)
Origin of circulating lineages (tMRCA))	19 February 2026 (95% HPD 04 January–21 March 2026)
Cases, 1 May – 5 Oct 2026	8,858 confirmed; 27.0% test positivity
Sequencing coverage vs. case burden	slope 0.71, R^2 = 0.55
Median age, female sex (%)	27 yr (IQR 16–39); 51.7% female

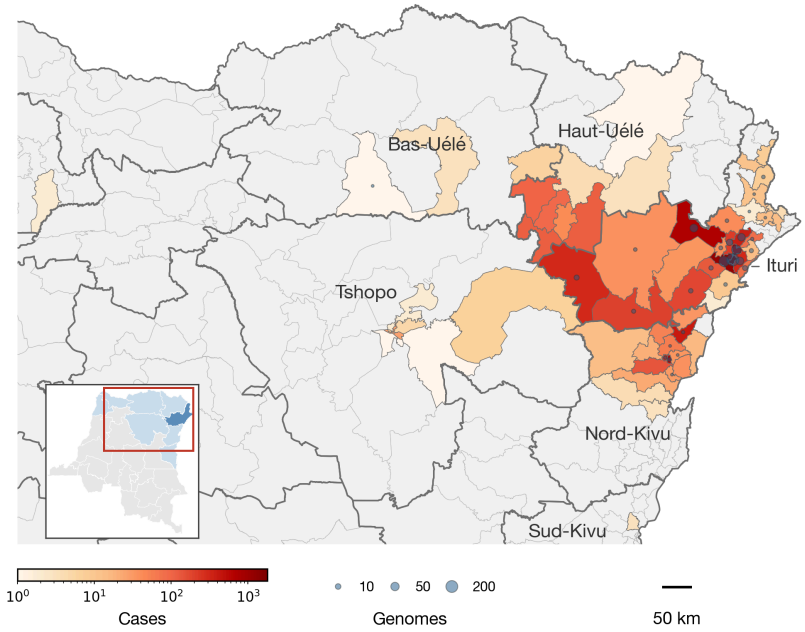


Figure 1. Health zones with confirmed cases and BDBV genomes through 5 October 2026. Health-zone fill shows case counts and overlaid circle size shows genome counts. Inset locates the affected provinces (Ituri in dark blue); the red outline marks the mapped area.

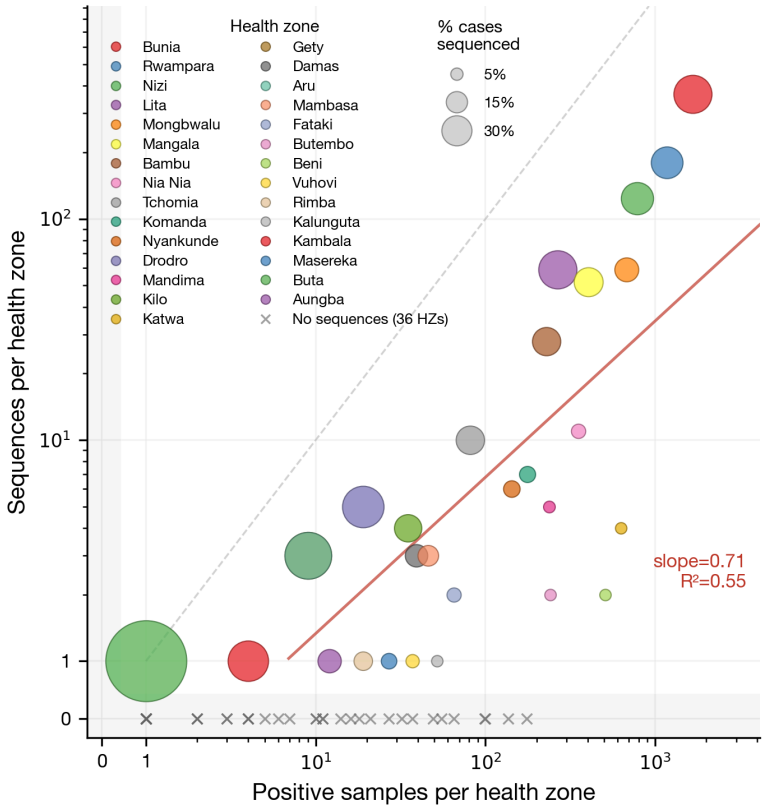


Figure 2. Genomic sampling intensity by health zone. Each circle is a health zone, sized by the percent of cases sequenced. The dashed grey unity line represents one sequence per positive sample; the red line is the regression fit across health zones with ≥1 sequence (slope 0.71, R^2 = 0.55). Health zones with positives but no sequences are marked "×".

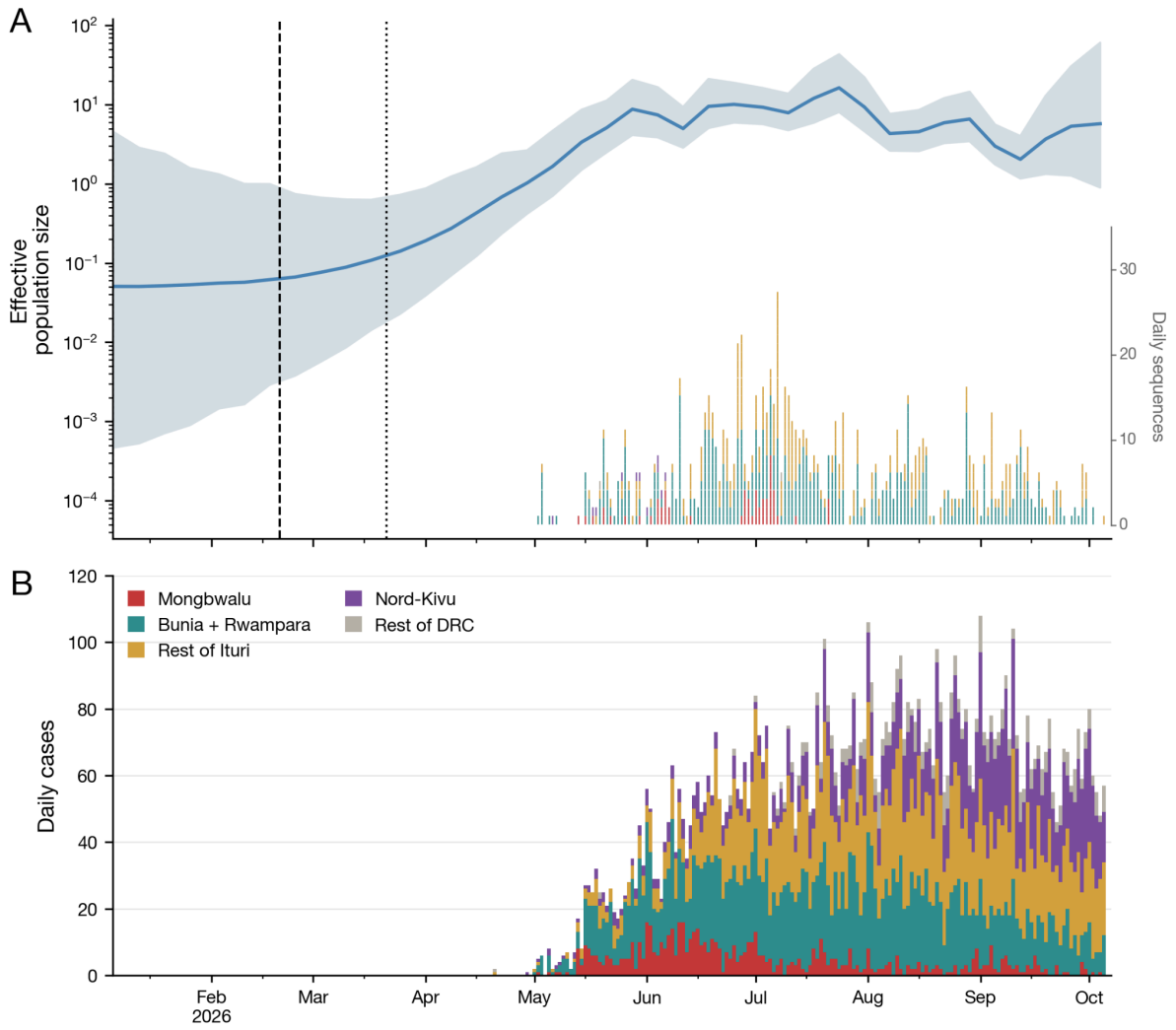


Figure 3. Evolutionary history of the early 2026 Bundibugyo virus disease outbreak in the DRC. a, SkyGrid reconstruction of relative virus effective population size, an indirect measure of infections, over time (blue line, mean; shading, 95% HPD). The dashed line marks the median tMRCA and the dotted line the upper bound of its 95% HPD; the lower bound coincides with the y-axis limit. The rug plot shows dates and frequency of included genomes, coloured as in panel b. b, Daily PCR-positive case counts by symptom onset in Mongbwalu (red), Bunia and Rwampara (green), the rest of Ituri (gold), Nord Kivu (purple), and the rest of the DRC (grey).

Implications and value of viral genome sequencing

- Continued effort to sequence samples from health zones newly reporting cases will help put their epidemics in the context of the wider epidemic.
- Sampling recent positive cases will allow continued situational awareness (e.g., is the epidemic growing or declining; are clusters of cases in one location related to frequent virus movements or a self-sustaining epidemic).
- Mandima, Beni, Katwa, Rwampara, Musienene, and Mambasa are currently undersampled with respect to the number of cases since the start of September and their recent growth rate (Figure 4). Oicha & Wamba have no sampled genomes, >70 cases, and consistent growth in the two weeks.
- Prioritising these areas for sequencing will provide better understanding of the drivers of the growth in cases they are experiencing.

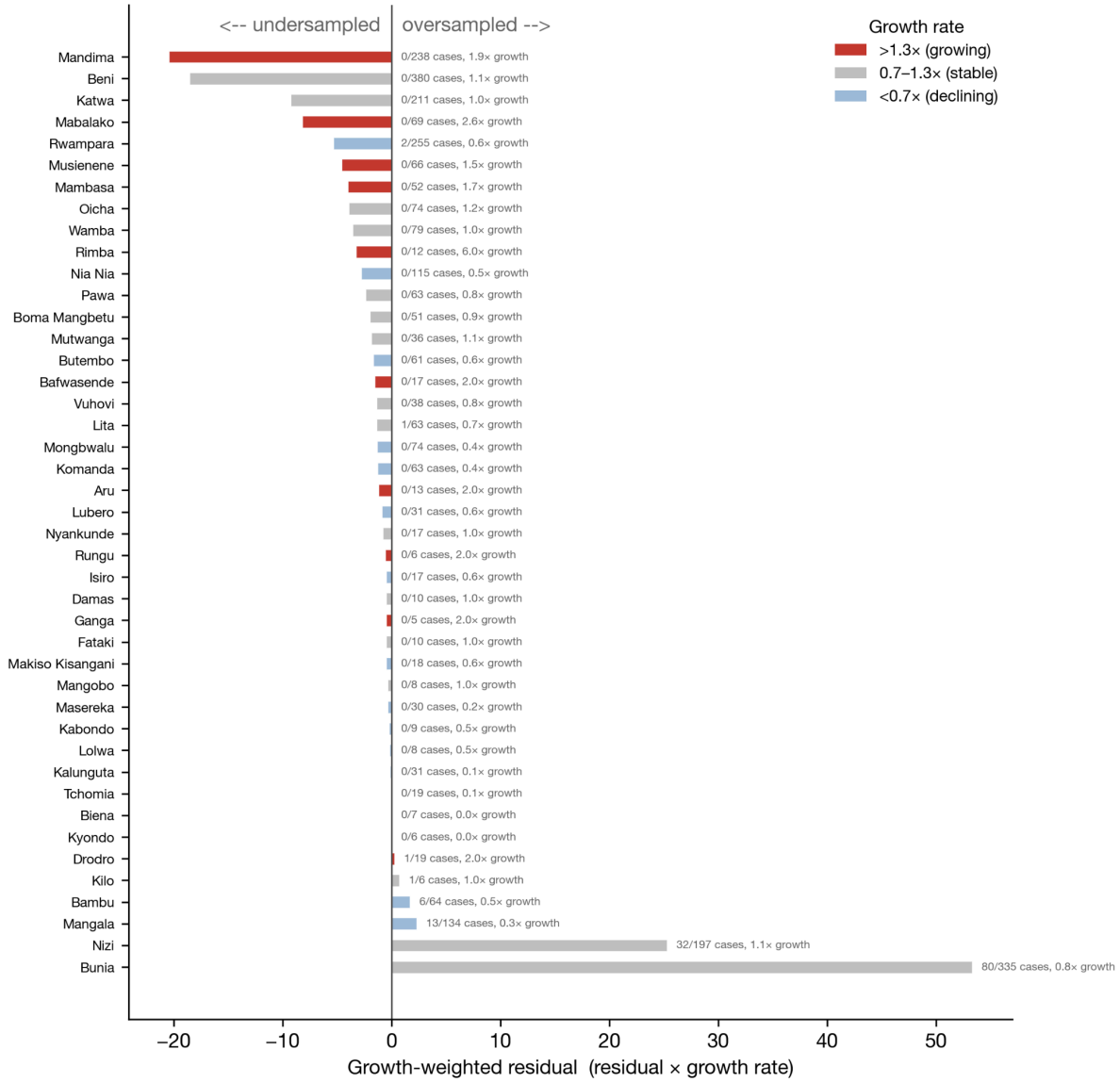


Figure 4. Recent genomic sampling intensity relative to number of cases and growth rate of the outbreak. The bars indicate the residual (actual number of genomes per health zone minus the number expected had genomes been distributed proportionally to cases) multiplied by growth rate (number of confirmed cases in the two weeks leading up to 5 October divided by the number in the preceding two weeks). The bars are colored by the growth rate, indicating a relatively declining, stable, or growing outbreak in that health zone.

Technical Appendix

1. Data and methods

- **Specimens.** EDTA-whole blood from living patients and oropharyngeal swabs from decedents, collected 1 May – 5 October 2026 in Ituri and Nord Kivu provinces as part of the emergency response (ethics approval ESP/CE/81/2026, Kinshasa School of Public Health).
- **Whole genome Sequencing.** Samples with Ct < 31 were sequenced using two approaches: early-phase samples used viral probe-based capture enrichment, subsequently by tiled-amplicon using ARTIC pan-Ebola or BDBV-specific primer panels. Sequencing libraries were run on an Illumina NextSeq 2000 or Oxford Nanopore GridION sequencer. Consensus genomes were generated via the ARTIC amplicon-nf pipeline (ONT) or Viralrecon 2.6.0 (Illumina). Median genome coverage 97.1% (IQR 92.0–97.2).
- **Quality control.** Genomes aligned to reference were downloaded from Pathoplexus (https://pathoplexus.org/). Three genomes appeared to have sequencing issues and were excluded. Host ADAR-editing signatures (clustered T→C substitutions) were detected with raccoon and masked — including 49 excess T→C sites in the glycoprotein — to prevent inflation of the inferred substitution rate. Iterative root-to-tip regression removed one strong temporal outlier, giving a final set of 982 genomes (971 Ituri, 10 Nord Kivu, and 1 Sud Kivu).
- **Phylogenetic inference and molecular dating.** Maximum-likelihood tree in IQ-TREE 3 (GTR+F+R3, selected by ModelFinder). Time-scaled inference in BEAST X v10.6.0-beta4 using a SkyGrid coalescent model (46 weekly transition points, 47-week cut-off), GTR substitution model and a strict molecular clock; 100 million MCMC generations, 10% burn-in, all relevant ESS > 100.
- **Epidemiological data.** Confirmed-case line list and weekly situation reports, 1 May – 5 October 2026. Missing symptom-onset dates were imputed from the fitted onset-to-collection delay distribution (epidist R package v0.4). A live dashboard is maintained at https://inrb-umie.github.io/BDBV2026-Epidemic_Dashboard/.

2. Additional details on main results

2.1 Epidemiological distribution (Figures 1,2)

Between 1 May and 5 October 2026, 8,858 PCR-confirmed cases of Bundibugyo virus disease were reported with an overall test positivity of 27.0%. Ituri province accounted for 6,512 cases (73.5%), concentrated in four health zones: Bunia (1,656; 18.7%), Rwampara (1,172; 13.2%), Mongbwalu (676; 7.6%) and Nizi (781; 8.8%). Sequencing efforts tracked case burden closely but sub-proportionally (slope 0.71, $R^2 = 0.55$ on the log-log scale), and 36 health zones with confirmed cases yielded no sequences at all — a clear geographic gap in genomic coverage. The correlation coefficient ($R^2 = 0.55$) has declined since the previous report, with more affected health zones lacking genomic coverage than before.

2.2 Timing of emergence and evolutionary rate (Figure 5)

The estimated evolutionary rate is 1.1×10^{-3} substitutions per site per year (95% HPD $1.0–1.2 \times 10^{-3}$), consistent with other orthoebolavirus outbreaks. Relative to our previous report (English | Français), we find a slightly earlier but broadly consistent tMRCA of mid-February, rather than early March, with a median tMRCA of 19 February 2026 (95% HPD 4 January – 21 March 2026). Because the tMRCA marks the common ancestor of sampled lineages, the true spillover may be earlier. We consider it possible that the virus may have been circulating in humans for roughly two to three months before the first laboratory confirmation.

2.3 Transmission dynamics (Figure 3)

The SkyGrid reconstruction shows three phases: a slow start in January–February 2026, marked acceleration from March to June, then stabilisation of effective population size through the end of the

study period. The effective population size is an indirect measure of incidence and is mirrored in the case data. Our previous report (English | Français) showed a decline in late August and early September 2026, which was likely reflective of the core part of the outbreak (*i.e.*, Mongbwalu, Bunia, and Rwampara) at that time experiencing a decline; since then the outbreak has spread to other regions and the effective population size estimates could represent an average of declining and growing outbreaks in different regions. Phylogenetic sub-clusters are not geographically structured within Ituri, indicating repeated movement of BDBV-infected individuals between health zones rather than contained local chains. Mongbwalu, Bunia, and Rwampara show a steady decline in daily case counts since July 2026.

2.4 Robustness and operational value of real-time genomics

Genome availability declined after mid-July 2026. Phylodynamic inference appears stable now, but the effective population size estimates could be the aggregation of both declining and growing outbreaks across the country.

3. Limitations and caveats

- Sampling is heavily Ituri-weighted. 971 of 982 analysed genomes are from Ituri, with ten genomes from Nord Kivu and one from Sud Kivu. Conclusions about spread to other provinces are not currently available from the genomic data.
- Estimates of population-size trajectories (Figure 2a) should be considered together with the presented uncertainty. This uncertainty becomes greater towards the present due to the delay between case detection and sequence availability.
- tMRCA is a lower bound on spillover timing. It dates the common ancestor of sampled lineages; unsampled early diversity would push the true introduction earlier.
- There are two data sources for case counts: INSP situation-reports and a line list. The line list is used for all analyses. If the line list included cases from health zones not documented in the INSP situation-reports, cases from those health zones are omitted. If the INSP situation-reports have cases from health zones not present in the line list, those cases and health zones are added to the line list. These differences and minor discrepancies in case counts reflect differing case-validation timelines.

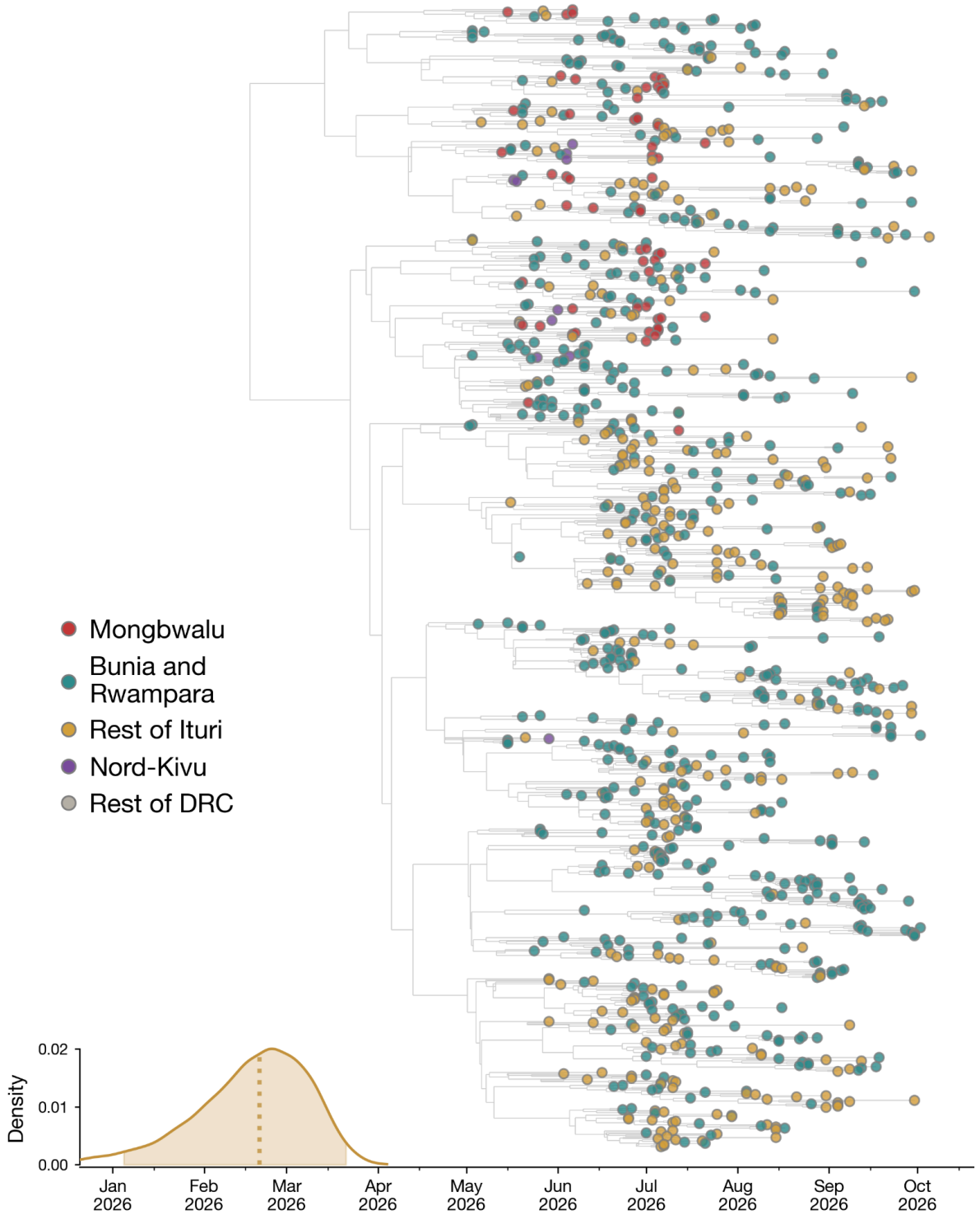


Figure 5. Time-resolved phylogeny of the 2026 Bundibugyo virus disease outbreak in the DRC. The density plot shows the time of the most recent common ancestor (tMRCA), with the dotted line indicating the median estimate and the shaded region representing the 95% HPDI.